CORY DIEHL

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OBJECTIVE

Develop and design professional systems in RF analog and digital electronics. Using applied math to develop automated tests and metrics in the verification process.

EDUCATION

California State University Sacramento
Bachelors in Electrical Engineering with a Minor in Math.
G.P.A. 3.9

EXPERIENCE

CSUS Power Generation Project, Sacramento CA — Summer 2018 - Spring 2019
Job: Research Assistant
- Helped design piezo-electric battery charging system and designed tests to meet project milestones. I was also in charge of analyzing the test data and training new team members.

Engineered Medical Technologies, Rocklin CA — Winter 2018
Job: Lab Tech
- Determined thermal uncertainty in the lab equipment by designing a PID controlled thermal plate for each test to be done on.

Sierra College Math Center, Rocklin CA — Fall 2014 - Spring 2017
Job: Math and Physics Tutor
- Helped students understand difficult concepts and misconceptions.

Programming

- LaTex, C, Python, Javascript, MatLab, Veralog, and VHDL

Relevant Courses

- Probability of Random Signals
- Digital Signal Processing
- Functional Analysis (Audited)
- Microwaves 1 & 2
- Modern Communication Systems

Microprocessors

- Microprocessors that I'm familiar with
  - BladeRF .... 1 year of experience
  - AVR Processors .... 2 years of experience
  - ARM Processors .... 2 years of experience
  - Xilinx Spartan 6 FPGA .... 1 year of experience
  - Cyclone IV,V, and 10 FPGA .... 1 year of experience
- IDEs that I'm familiar with
  - Atmel Studio
  - Code Composer Studio
  - MPLab X
  - Quartus
  - GNU Radio
  - Xilinx ISE

Projects

- Combated emergency services inter agency interoperability communications issues using GNU Radio and a BladeRF software defined radio with a Cyclone V FPGA. (Senior Project)
- Made a wireless home lighting system with the ESP8266.
- Presented Research at Sonoma State Undergraduate Math Conference Fall 2018
- Editor of the published real analysis text book ”Real Analysis: A Long-Form Mathematics Textbook” by Dr. Cummings. The book is on Amazon, and currently used at CSUS and other universities.